FP8 Grouped GEMM

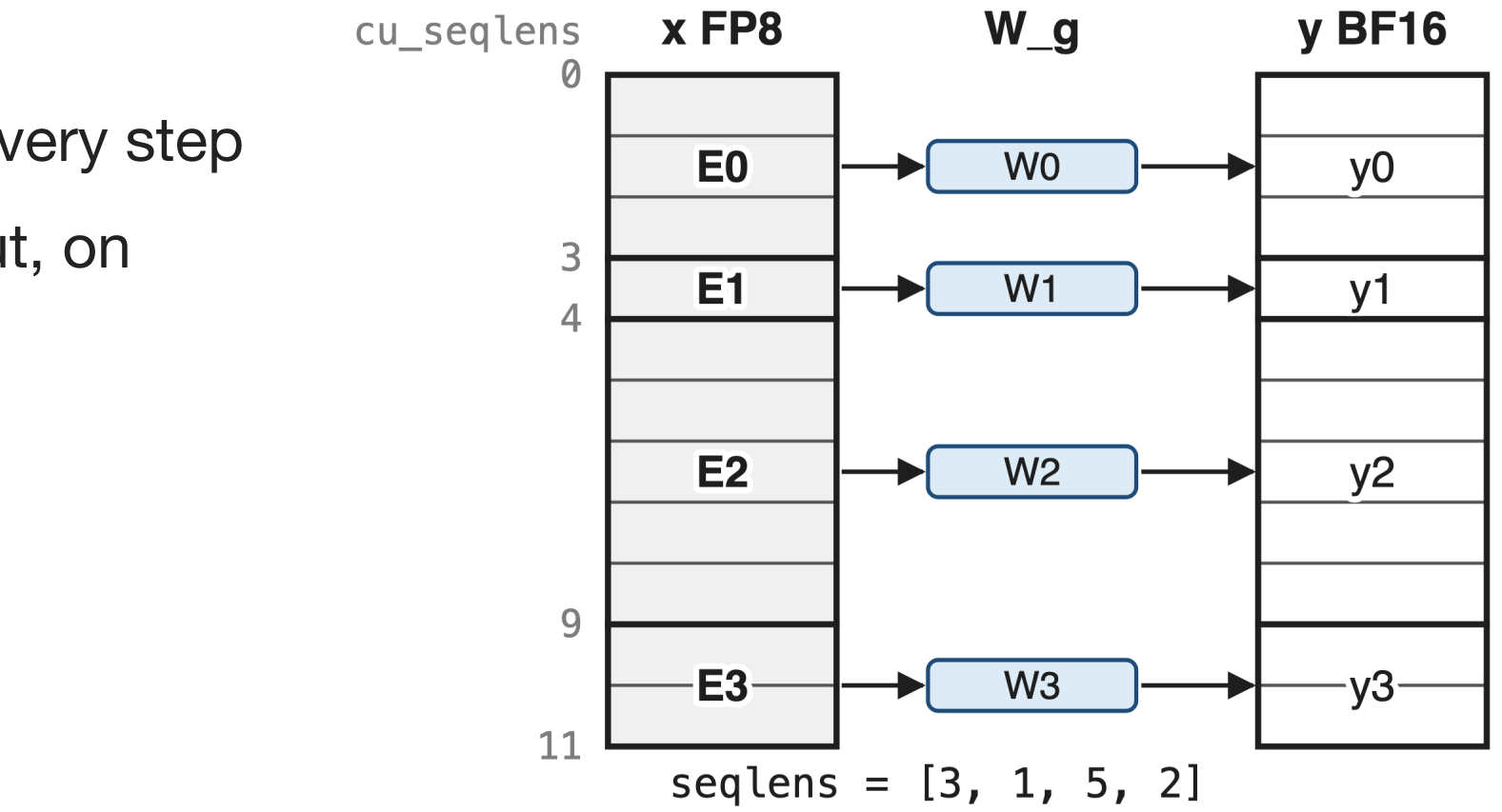
All experts' FP8 matmuls in one launch, any token split

Tencent Hunyuan AI Infra · HPC-Ops

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

MoE layers multiply every expert's tokens at once

- One group per expert: $y_g = x_g \cdot W_g^T$
- Group sizes change every step
- FP8 e4m3 in, BF16 out, on H20 (SM90)



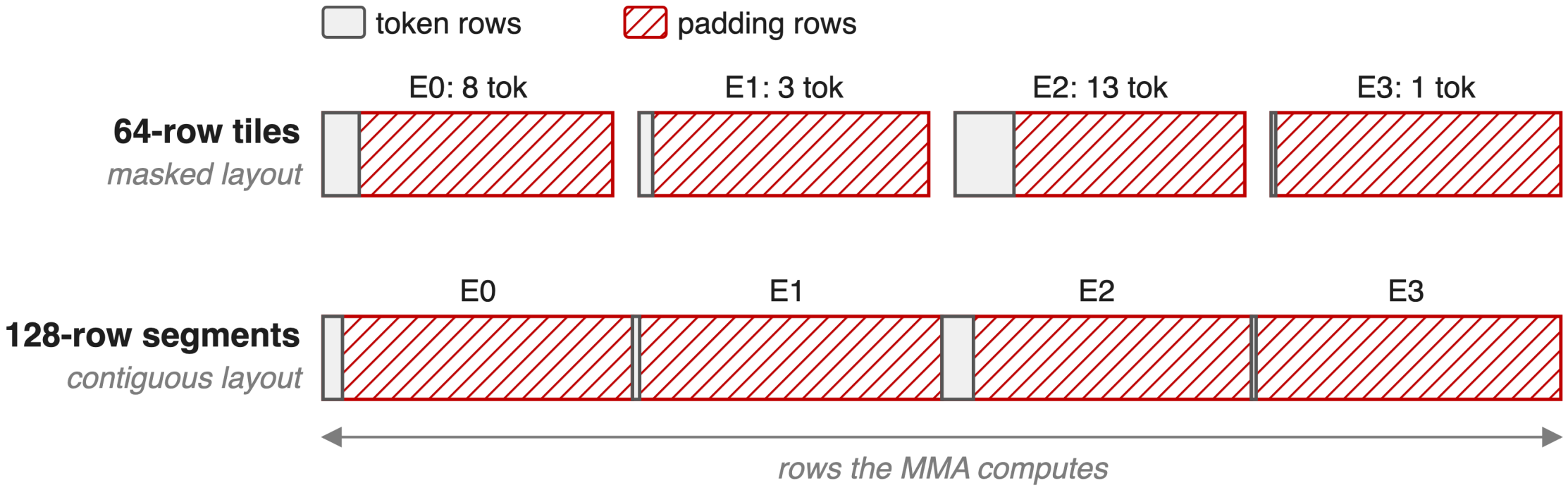
Decode experts fill one-eighth of a tile

87.5%

of a 64-row MMA tile is padding when an expert receives 8 decode tokens (analytic)

Hopper FP8 WGMMA atoms are 64×N×32 (CUTLASS cute SM90 GMMA); DeepGEMM SM90 grouped BLOCK_M ≥ 64; 8 tokens per group as in the HPC-Ops README example

Token-major tiles pad every small expert



The contiguous layout also forces each expert onto a 128-row boundary

Insight: put tokens on the MMA's narrow side

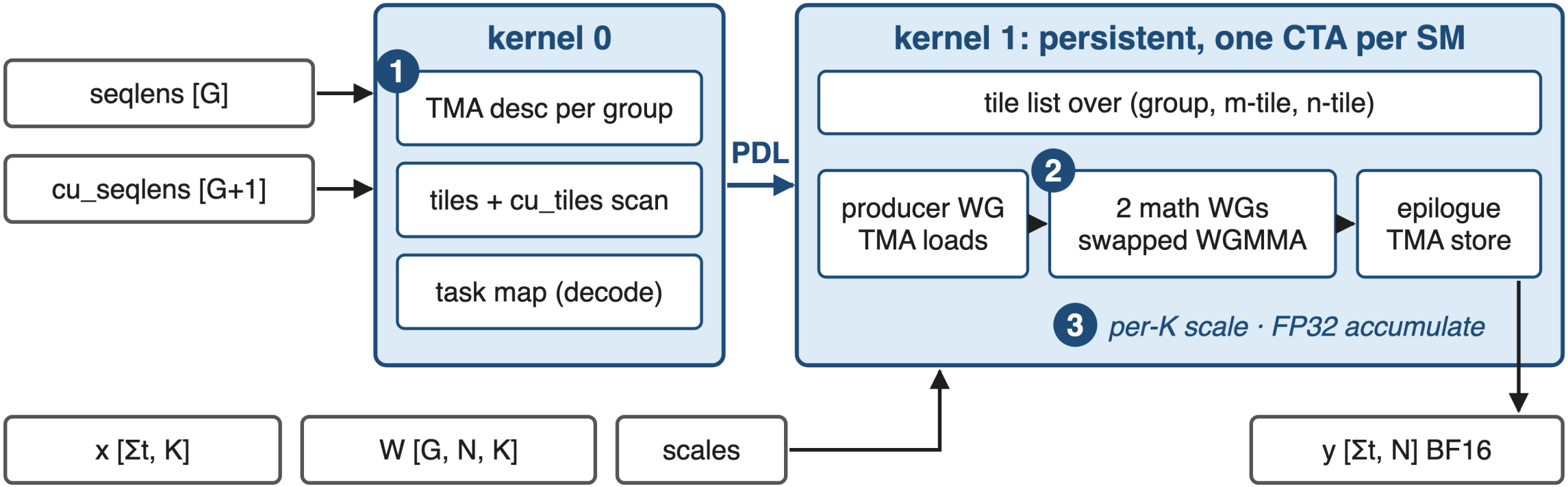
- Hopper FP8 MMA: M fixed at **64**; N from **8** to **256** in steps of 8
- Weights always fill 64 rows; decode tokens rarely do

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- Hopper FP8 MMA: M fixed at **64**; N from **8** to **256** in steps of 8
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Compute $y^T = W \cdot x^T$: weights take M, tokens take N, and the tile shrinks to 8

Kernel 0 describes groups, kernel 1 streams tiles

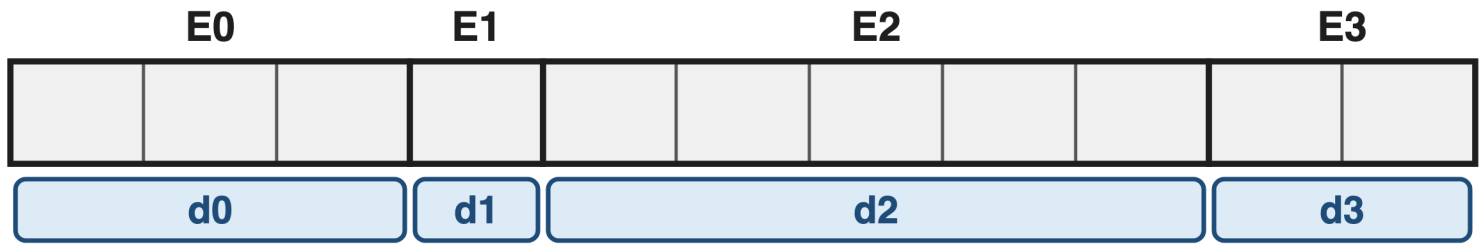


Same structure for per-tensor and block-wise scaling; the per-tensor path launches kernel 1 with PDL

❶ cu_seqlens become per-group TMA descriptors

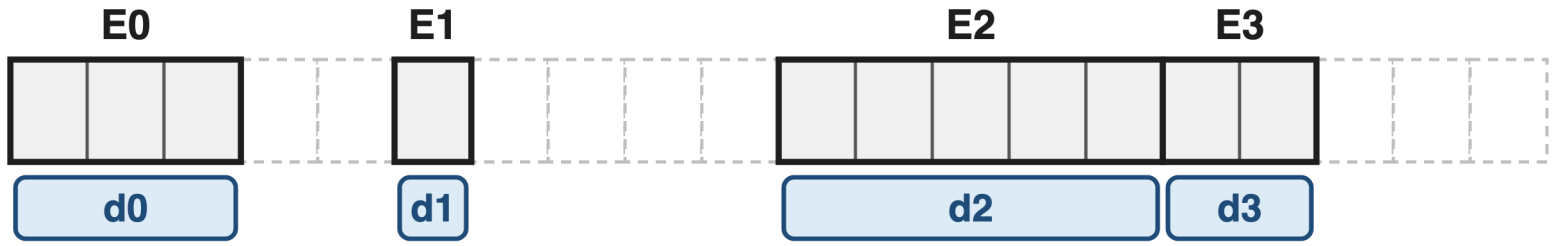
compact

`cu = 0 3 4 9 11`



DeepEP layout

`cu = 0 5 10 15`

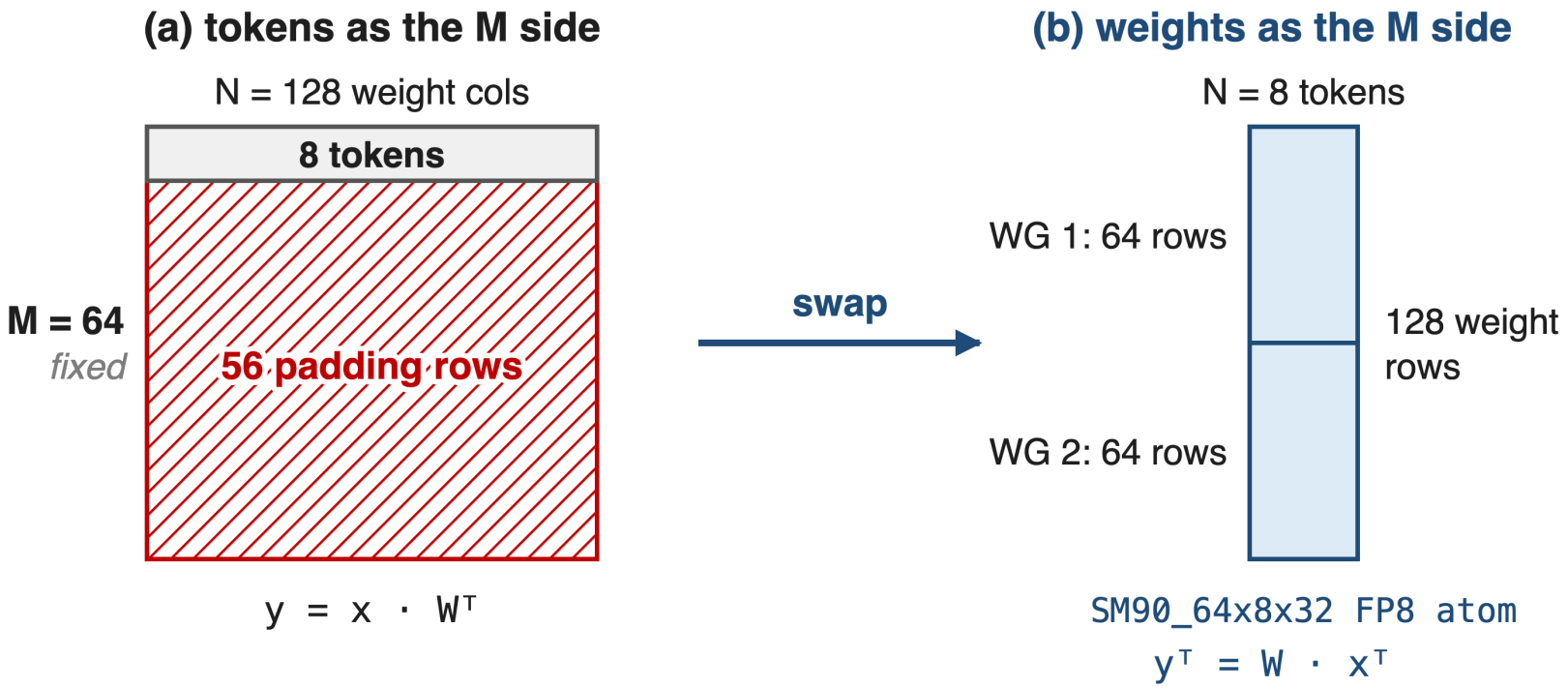


TMA descriptor: base and rows of one group

never loaded

No padding copy, no host sync: seqlens = how many, cu_seqlens = where

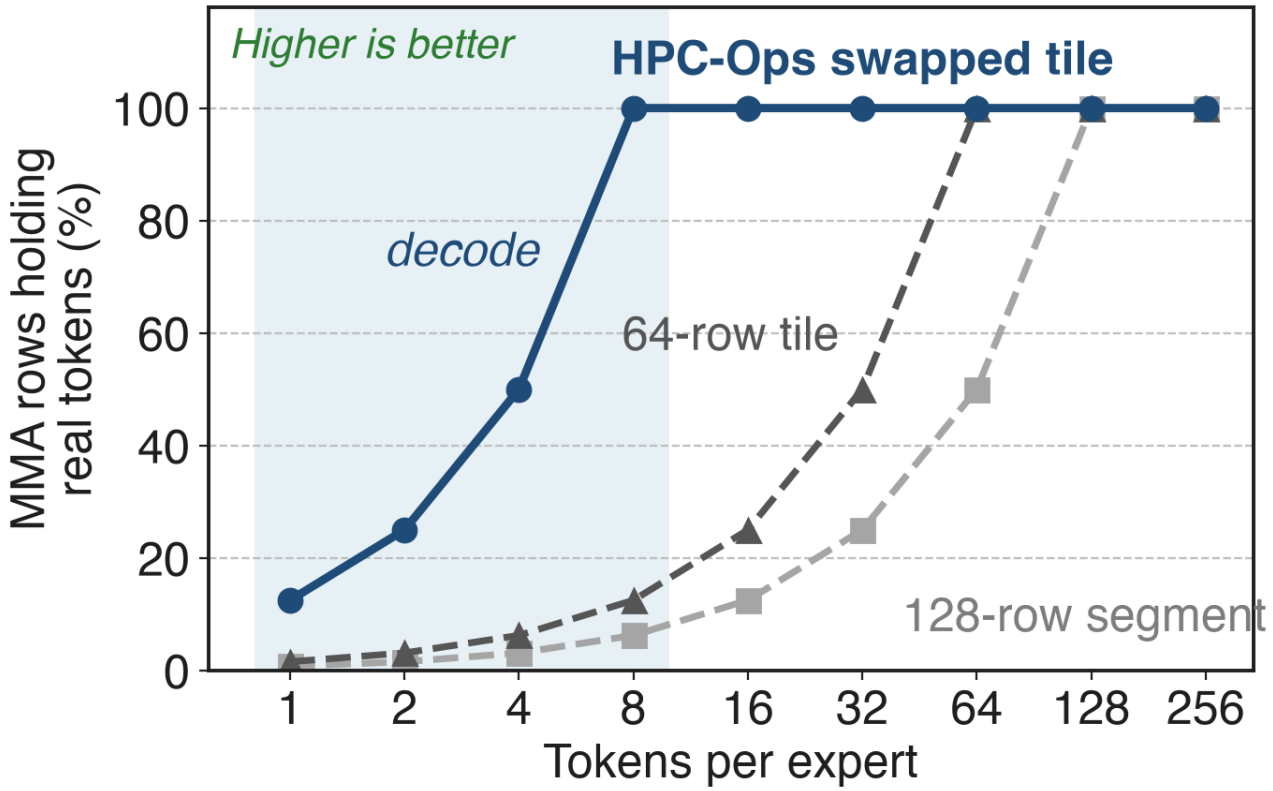
❷ Swapping operands lets the tile shrink to 8



Token tile follows the average group size: 8, 16, 32, 48 or 64

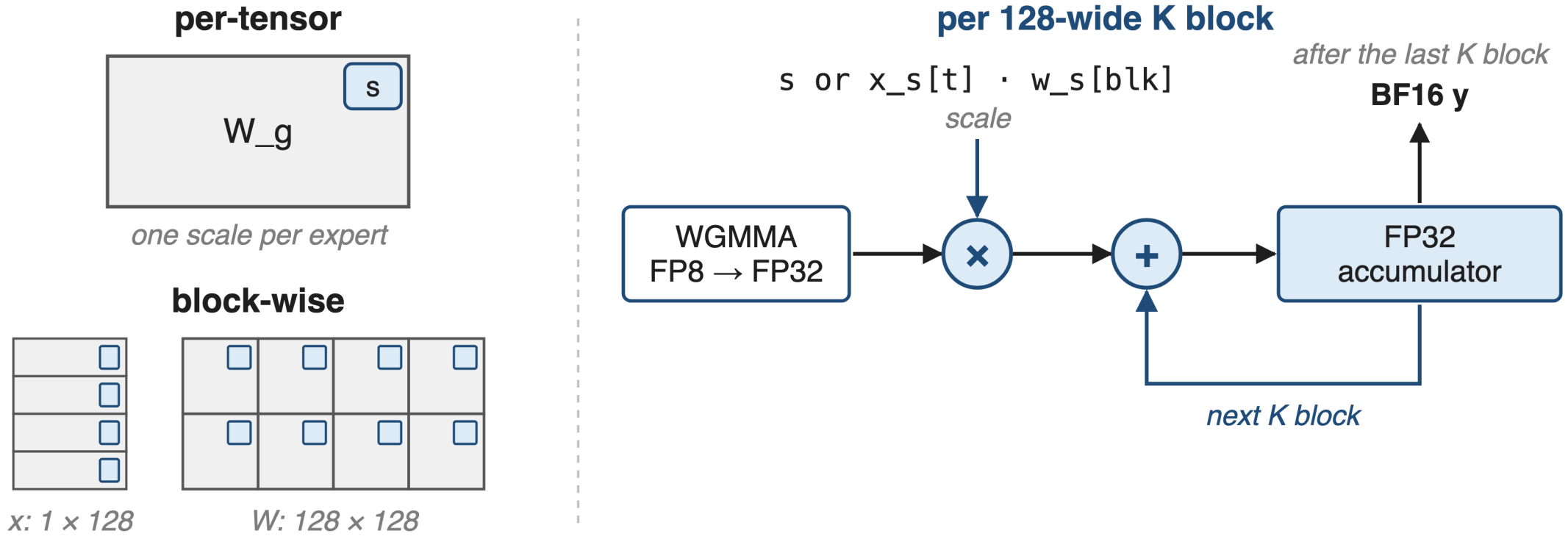
At decode sizes, 8× more MMA rows do work

- 8 tokens: **100%** vs **12.5%**
- 1–4 tokens: still **8×** a 64-row tile
- Equal from 64 tokens up



Analytic, uniform group sizes; tile rules from HPC-Ops and DeepGEMM source

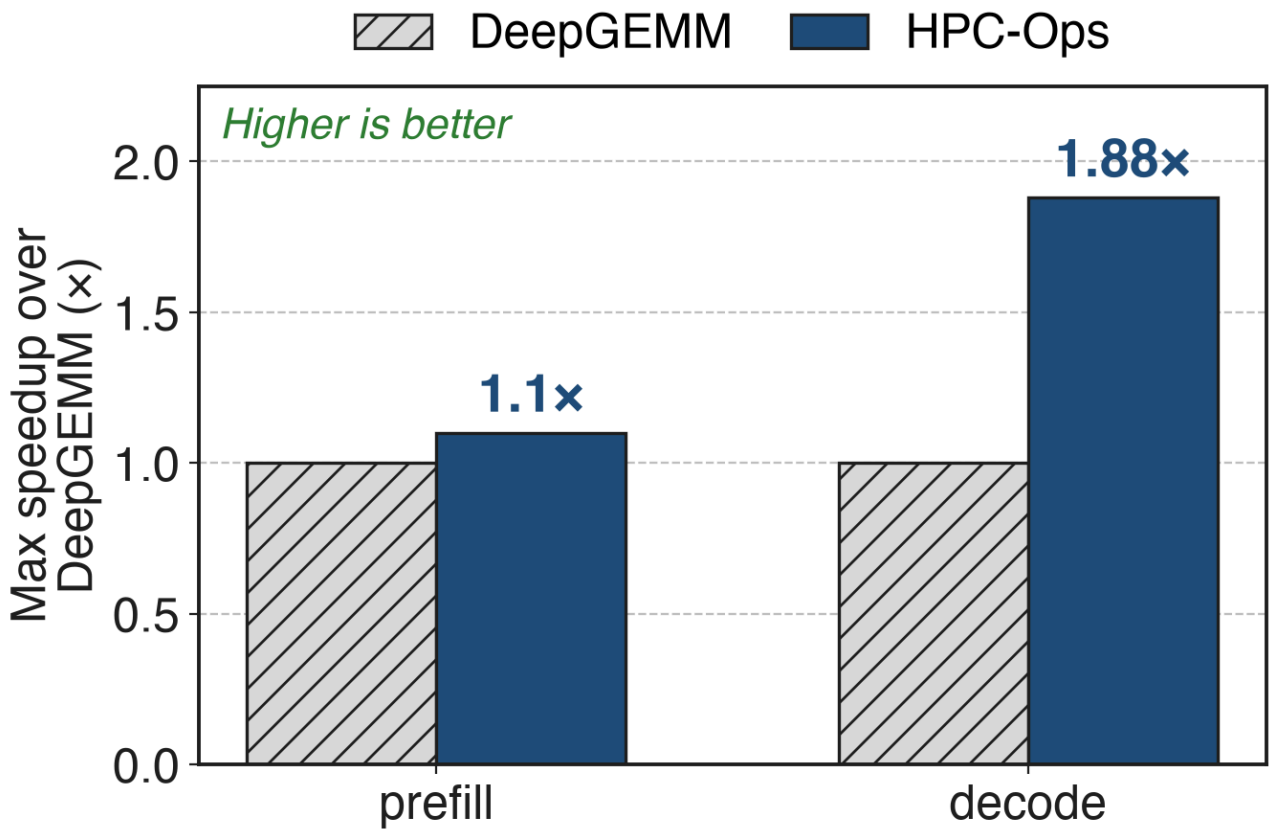
❸ One mainloop handles both FP8 scalings



Every 128-wide K block is scaled and promoted into FP32 registers

Up to 1.88× over DeepGEMM at decode

- Prefill: up to **1.1×**
- Expert grouped matmul, H20
- Consistent with the tile analysis: gains where groups are small



Maximum observed speedups as published in the HPC-Ops README; per-shape data not released

The price: Hopper only, one shape per call

- SM90 only, tuned on H20
- Same N and K for every group
- One tile size from the average

- Block-wise: x_scale transposed and padded per group
- DeepEP input: reformat, K = **2048 / 4096 / 7168**
- Max-only results published

Summary

- Decode experts get few tokens: a 64-row tile is **87.5%** padding at 8
- **Insight**: swap operands so tokens take the MMA's 8-wide side
- cu_seqlens → per-group TMA descriptors; per-tensor and block-wise FP8
- Up to **1.88×** (decode) and **1.1×** (prefill) over DeepGEMM on H20

https://github.com/Tencent/hpc-ops